

LINEAR DEPENDENCE & INDEPENDENCE OF VECTOR:

Let the set of vectors $V_1, V_2, \dots, V_n$ the vectors can be check for linear dependence & independence by the below condition:

$$C_1V_1 + C_2V_2 + \dots + C_nV_n = 0$$

1) For the vector's equation

$C_1V_1 + C_2V_2 + \dots + C_nV_n = 0$ & write a homogeneous system of linear equations in the terms of $C_1, C_2, \dots, C_n$.

2) By applying the Gauss-Jordan Elimination method to find the constant then two condition will arise:

a) If the system of linear equations has non-trivial solution means $(C_1 \neq 0, C_2 \neq 0, \dots, C_n \neq 0)$ then set of vectors linearly dependent.

b) If the system of linear equation has trivial solution means $(C_1 = 0, C_2 = 0, \dots, C_n = 0)$ then set of vectors linearly independent.

Example: Determine whether the set of vectors in $\mathbb{R}^3$ is linearly dependent or linearly independent given by.

$$S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}.$$

Form the vector equation.

$$\therefore C_1 S_1 + C_2 S_2 + C_3 S_3 = 0$$

$$\Rightarrow C_1(1, 2, 3) + C_2(0, 1, 2) + C_3(-2, 0, 1).$$

$$\Rightarrow (C_1, 2C_1, 3C_1) + (0, C_2, 2C_2) + (-2C_3, 0, C_3).$$

$$\Rightarrow (C_1 - 2C_3, 2C_1 + C_2, 3C_1 + 2C_2 + C_3) = (0, 0, 0)$$

$$\Rightarrow C_1 - 2C_3 = 0$$

$$\Rightarrow 2C_1 + C_2 = 0$$

$$\Rightarrow 3C_1 + 2C_2 + C_3 = 0$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -2 & : & 0 \\ 2 & 1 & 0 & : & 0 \\ 3 & 2 & 1 & : & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 2 & 1 \end{bmatrix} \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & -1 \end{bmatrix} R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_1 - 2C_3 = 0 \text{ --- (ii)}$$

$$C_2 + 4C_3 = 0 \text{ --- (i)}$$

$$C_3 = 0$$

Putting C_3 in eq (i)

$$C_2 + 4(0) = 0$$

$$C_2 = 0.$$

Putting values of C_3 ~~& C_2~~ in (ii)

$$C_1 - 2(0) = 0$$

$$C_1 = 0.$$

This shows that the set of vectors are linearly independent.

NOTE:

- 1) Set of vectors are said to be linearly dependent if $|A| = 0$.
- 2) Set of vectors are said to be linearly independent if $|A| \neq 0$.

Example: Prove that the set of vectors
 $S = \{(1, -1, 2), (2, 3, -4), (0, 5, -8)\}$
are linearly independent.

$$\therefore C_1 S_1 + C_2 S_2 + C_3 S_3 = 0$$

$$\Rightarrow C_1(1, -1, 2) + C_2(2, 3, -4) + C_3(0, 5, -8) = 0$$

$$\Rightarrow (C_1, -C_1, 2C_1) + (2C_2, 3C_2, -4C_2) + (0, 5C_3, -8C_3) = 0$$

$$\Rightarrow (C_1 + 2C_2, -C_1 + 3C_2 + 5C_3, 2C_1 - 4C_2 - 8C_3) = (0, 0, 0)$$

$$\Rightarrow C_1 + 2C_2 = 0$$

$$\Rightarrow -C_1 + 3C_2 + 5C_3 = 0$$

$$\Rightarrow 2C_1 - 4C_2 - 8C_3 = 0$$

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 5 \\ 2 & -4 & -8 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 5 \\ 2 & -4 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 5 & 5 \\ 0 & -8 & -8 \end{bmatrix} \begin{array}{l} \\ R_2 + R_1 \\ R_3 - 2R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{array}{l} \\ R_2 \div 5 \\ R_3 \div -8 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} R_3 - R_2$$

$$C_1 + 2C_2 = 0 \dots (ii)$$

$$C_2 + C_3 = 0 \dots (i)$$

From eq (i) $\Rightarrow C_2 = -C_3$

Putting C_2 in eq (ii) $\Rightarrow C_1 = 2C_3$

$$C_2 = -C_3$$

$$C_3 = C_3$$

let $c_3 = a$; where a is real number

Here c_1, c_2 & c_3 has infinite values
so the given set of vectors are
linearly dependent.

SPANNING OF VECTORS:

If $\vec{u} = \{v_1, v_2, v_3, \dots, v_n\}$ is a set of vectors in a vector space V , then the span of V is the set of all linear combinations of the vectors in V is called spanning of a set. Formula can be use to span the set.

$$\text{Span}(V) = C_1 v_1 + C_2 v_2 + C_3 v_3 + \dots + C_n v_n.$$

PROCEDURE:

STEP 1: Let an arbitrary vector u .

STEP 2: Apply the condition for spanning.

$$u = C_1 v_1 + C_2 v_2 + \dots + C_n v_n.$$

STEP 3: Decision

• If $|A| \neq 0$ then system has solutions & vectors can be expressed for linear combination.

• If $|A| = 0$ then system has no solutions & vectors can not be expressed for linear combination.

Show that the set of vectors
 $V = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$ span in $\mathbb{R}^3$.

let $u = (u_1, u_2, u_3)$

$$\therefore u = C_1 V_1 + C_2 V_2 + C_3 V_3 + \dots + C_n V_n.$$

$$\Rightarrow (u_1, u_2, u_3) = C_1(1, 2, 3) + C_2(0, 1, 2) + C_3(-2, 0, 1).$$

$$\Rightarrow (u_1, u_2, u_3) = (C_1 - 2C_3, 2C_1 + C_2, 3C_1 + 2C_2 + C_3).$$

$$\Rightarrow C_1 - 2C_3 = u_1.$$

$$\Rightarrow 2C_1 + C_2 = u_2.$$

$$\Rightarrow 3C_1 + 2C_2 + C_3 = u_3.$$

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow |A| = 1 \begin{vmatrix} 1 & 0 & -2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} = 1(1(1-0) - 0 - 2(4-3))$$

$$\Rightarrow |A| = 1(1-0) - 0 - 2(4-3)$$

$$\Rightarrow |A| = 1(1) - 0 - 2(1)$$

$$\Rightarrow |A| = 1 - 0 - 2$$

$$\Rightarrow |A| = -1. \quad \Rightarrow |A| \neq 0$$

So, set of vectors V_1, V_2 & V_3 can be expressed by linear combination. Hence, Set of vectors is Spanning in $\mathbb{R}^3$.

BASIS IN VECTOR SPACE

A set of vectors $\vec{V} = \{V_1, V_2, \dots, V_n\}$ in a vector space is said to have basis for V if it satisfies two conditions.

- 1) Set of vectors $\vec{V} = \{V_1, V_2, \dots, V_n\}$ must be span.
- 2) Set of vectors $\vec{V} = \{V_1, V_2, \dots, V_n\}$ must be linearly independent.

Show that the following set is a basis in $\mathbb{R}^3$.

$$V = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

If $V = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is the basis in $\mathbb{R}^3$ then it must satisfy two conditions:

1) Condition: (i) set of vectors $V = \{V_1, V_2, \dots, V_n\}$ must be span. Here

$$U = C_1 V_1 + C_2 V_2 + C_3 V_3 + \dots + C_n V_n.$$

$$\Rightarrow (u_1, u_2, u_3) = C_1(1, 0, 0) + C_2(0, 1, 0) + C_3(0, 0, 1)$$

$$\Rightarrow (u_1, u_2, u_3) = (C_1, C_2, C_3).$$

$$\Rightarrow C_1 = u_1$$

$$\Rightarrow C_2 = u_2$$

$$\Rightarrow C_3 = u_3$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Here } |A| = 1 \neq 0$$

Condition (ii): Set of vectors.

$V = \{V_1, V_2, \dots, V_n\}$ must be linearly independent

$$C_1 V_1 + C_2 V_2 + C_3 V_3 + \dots + C_n V_n = 0$$

$$\Rightarrow C_1(1, 0, 0) + C_2(0, 1, 0) + C_3(0, 0, 1) = (0, 0, 0)$$

$$\Rightarrow (C_1, C_2, C_3) = (0, 0, 0)$$

$$\Rightarrow C_1 = 0$$

$$\Rightarrow C_2 = 0$$

$$\Rightarrow C_3 = 0$$

We get the trivial solution $C_1 = 0, C_2 = 0$ & $C_3 = 0$. So, the set of vectors is linearly independent. As both conditions are satisfied hence set of vectors is the basis for $\mathbb{R}^3$.

NULLITY:

A matrix that satisfies the condition
Nullity of matrix = (number of unknown)
- Rank of matrix.

Q: Find the nullity of matrix:

$$A = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 \\ -2 & -1 & 1 & -1 & 3 \\ 0 & 3 & 9 & 0 & 12 \end{bmatrix}$$

Using elementary row operation (echelon form)

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & -1 & -3 & 1 & 3 \\ 0 & 3 & 9 & 0 & 12 \end{bmatrix} \begin{array}{l} \\ R_2 \times (-1) \\ R_3 + 2R_1 \\ \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 21 \end{bmatrix} \begin{array}{l} \\ \\ R_3 + R_2 \\ R_4 - 3R_2 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & -1 & -3 \\ 0 & 0 & 0 & 3 & 21 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{Swapping } R_3 \text{ by } R_4$$

Rank of $A = 3$.

$$\text{Nullity} = 5 - 3 = 2.$$